

Physiological Characteristics of *Prunus×yedoensis* Matsum. in Street Trees in Goyang and Paju, South Korea

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Abstract

Prunus×yedoensis Matsum. is a commonly planted street tree species in South Korea. In this study, we assessed the physiological characteristics of *P.×yedoensis* growing along streets in Goyang and Paju, located in Gyeonggi Province. Across the surveyed sites, net CO₂ assimilation rates ranged from 15.8 to 18.1 μmol CO₂ m⁻² · s⁻¹, chlorophyll fluorescence (Fv/Fm) values ranged from 0.81 to 0.85, and proline contents were between 3.4 and 3.7 mg · g⁻¹ fresh weight. While the trees appeared to be under some degree of physiological stress, the levels of chlorophyll fluorescence and proline content suggested that the stress was not severe. These findings indicate that *P.×yedoensis* street trees in the studied areas are not experiencing significant physiological decline. Continued monitoring will be essential for evaluating the long-term physiological responses of urban trees under environmental stress.

Key Words: street tree, urban trees, photosynthesis, physiological characteristics, tree health monitoring

Introduction

Prunus×yedoensis Matsum. a member of the Rosaceae family and the Amygdaloideae subfamily, is a species of cherry tree primarily found in tropical regions of Asia and the Americas. Approximately 200 species within this genus have been identified worldwide (Kalkman 2004; Han et al. 2022). *P.×yedoensis* is widely planted as a street tree due to its ornamental appeal, strong tolerance to air pollution, and remarkable adaptability to adverse environmental conditions. The area covered by this species in urban forests is expected to continue expanding (Han et al. 2019; Oh et al. 2020;

Ha et al. 2022).

In recent years, the introduction of certified arborists has increased the demand for professional and systematic evaluations of pest infestations and the physiological status of urban trees. However, accurate diagnosis remains challenging (Lee et al. 2022b). In particular, under drought or other abiotic stresses, tree health is often assessed based on visual inspection and empirical guidelines (Boone and Westwood 2006). When expertise or relevant data is lacking, it becomes difficult to accurately evaluate the physiological characteristics of trees. This problem is especially pronounced in mature trees, for which no established standards

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or reference values currently exist.

Previous research on *P×yedoensis* has primarily focused on seasonal variations in leaf microstructure and particulate matter adsorption in urban trees in southern regions (Jin et al. 2021), as well as surveys of pests and diseases (Kim et al. 2009; Ahn et al. 2022). Studies have also examined the tree's physiological responses to air pollutants (Lee et al. 1999), heavy metal contamination (Kim and Song 2015; Park et al. 2021a), and calcium chloride treatment during winter (Je and Kim 2016). However, most of these studies have been conducted on young saplings (2–3 years old), with relatively few investigations targeting the physiological traits of mature street trees.

This study aims to evaluate the physiological characteristics of *P×yedoensis* street trees in relation to the timing of urban development. In the context of ongoing climate change, establishing research and continuous monitoring systems for urban tree physiology is essential. These efforts will support the development of urban forests and street-scapes tailored to the environmental and social characteristics of specific regions, ultimately contributing to healthier and more sustainable urban ecosystems.

Materials and Methods

Study sites

This study selected *P×yedoensis*, a species commonly planted in urban areas and particularly favored as a street tree, as the subject of research. The study sites were in Goyang-si, Gyeonggi-do, specifically in Ilsanseo-gu near Kintex-ro (PY1, 4.7 km), the area around Soejae-ro; Paju Unjeong District (PY2, 1.4 km) developed since 2004, and the Mugunghwa-ro (PY3, 4.7 km) in Ilsandong-gu, Goyang-si, developed since 2010, including cultural complex facilities such as Kintex and Ilsan Lake Park (Table 1). The ambient air quality data (PM_{2.5}, PM₁₀, O₃, NO₂, CO and SO₂) were obtained from the official website of the Air Gyeonggi-Do Atmospheric Environment Information System (<https://air.gg.go.kr>), based on the monitoring results from the station closest to the study area (Table 2). All measurements were conducted between 8:00 am and 11:00 am on a single day in July 2021, under conditions without visible leaf damage. Additionally, six street trees were selected from survey sites, and measurements were repeated three times. To assess the growth condition of *P×yedoensis*, we recorded the DBH, soil moisture content (Spectrum[®] Field Scout, TDR 200, USA), and tree vitality (Junsmeter, PrumBio, Korea).

Table 1. Study area of *P×yedoensis* street trees in Goyang and Paju, Gyeonggi Province, South Korea

Sites	City	GPS
PY1	Goyang (Ilsandong-gu)	Kintex-ro: 37°39'42.1"N, 126°44'45.0"E
PY2	Paju (Un-jung)	Soejae-ro: 37°44'57.0"N, 126°46'27.3"E
PY3	Goyang (Ilsandong-gu)	Mugunghwa-ro: 37°40'34.9"N, 126°47'48.1"E

Table 2. Ambient air quality parameters measured in Goyang and Paju, Gyeonggi Province, South Korea

Sites	Air quality parameter					
	PM _{2.5}	PM ₁₀	O ₃	NO ₂	CO	SO ₂
	(μg · m ⁻³)		(ppm)			
PY1	20	27	0.042	0.013	0.4	0.003
PY2	19	29	0.042	0.011	0.5	0.002
PY3	12	41	0.040	0.015	0.3	0.003

PM_{2.5}, particulate matter 2.5; PM₁₀, particulate matter 10; O₃, ozone; NO₂, nitrogen dioxide; CO, carbon monoxide; SO₂, sulfur dioxide. PY1, Kintex-ro (Goyang); PY2, Soejae-ro (Paju); PY3, Mugunghwa-ro (Goyang).

Measurement of physiological characteristics

Photosynthesis

Photosynthetic measurements were conducted on the 3rd to 5th fully expanded leaves from the shoot apex, selecting only healthy leaves without visible damage. Measurements were carried out between 8:00 am and 11:00 am under controlled light conditions using a PLU5 LED light source (ADC BioScientific Ltd., UK) that provide a constant photosynthetically active radiation (PAR) level of $1,500 \mu\text{mol m}^{-2} \cdot \text{s}^{-1}$. Leaf gas exchange parameters were recorded with an LCI-SD Ultra Compact Photosynthesis System (ADC BioScientific Ltd., UK) including the net CO_2 assimilation rate (A), stomatal conductance (g_s), transpiration rate (E), intercellular CO_2 concentration (C_i), and water use efficiency (WUE). Water use efficiency was calculated as the ratio of the net CO_2 assimilation rate to the transpiration rate ($\text{WUE} = \text{A}/\text{E}$), following the methods described by Ashraf (2002) and Wang et al. (2001).

Chlorophyll fluorescence response

Chlorophyll fluorescence was measured using a continuous source chlorophyll fluorometer (OS-30P, ADC Bioscientific Ltd., UK). For measurement, sample clips were attached to leaf segments from the 3rd to 5th fully expanded leaves from the apex. Leaves were dark-adapted for 20 minutes to ensure stabilization of Photosystem II prior to measurement. Following dark adaption, chlorophyll fluorescence responses were measured (Maxwell and Johnson 2000). In chlorophyll fluorescence, F_v/F_m represents the maximum quantum yield of Photosystem II. The initial fluorescence (F_o), maximum fluorescence (F_m), and the maximum change in fluorescence ($F_v = F_m - F_o$) were calculated, and the photochemical efficiency ($F_v/F_m = [F_m - F_o]/F_o$) was compared.

Chlorophyll and proline content

Chlorophyll content was measured using fully expanded, healthy leaves collected from the 3rd to 5th positions below the shoot apex. Leaf discs (0.1 g), excluding major veins, were immersed in 10 mL of 80% (v/v) acetone. The samples were incubated in the dark at 4°C for one week to extract chlorophyll pigments. The absorbance of the extracts was measured at 470 nm, 645 nm, and 663 nm using a

UV/VIS spectrophotometer (UV-2100, Shimadzu, Japan). Based on the absorbance values, the concentrations of chlorophyll a, chlorophyll b, and total chlorophyll were calculated using the equations proposed by Arnon (1949) and Lichtenthaler (1987).

$$\text{Chlorophyll a} = 12.7 \cdot A_{663 \text{ nm}} - 2.69 \cdot A_{645 \text{ nm}}$$

$$\text{Chlorophyll b} = 22.9 \cdot A_{645 \text{ nm}} - 4.68 \cdot A_{663 \text{ nm}}$$

$$\text{Total chlorophyll (a+b)} = 20.2 \cdot A_{645 \text{ nm}} + 8.02 \cdot A_{663 \text{ nm}}$$

$$\text{Carotenoid} = (1,000 \cdot A_{470 \text{ nm}} - 1.82 \cdot \text{Chlorophyll a} - 85.02 \cdot \text{Chlorophyll b})/198$$

Proline content was measured using fully expanded leaves from the 3rd to 4th positions below the growing apex. A 0.5 g leaf sample was homogenized in 10 mL of 3% (v/v) sulfosalicylic acid. The homogenate was centrifuged at $3,400 \times g$ for 20 minutes at 4°C , and the resulting supernatant was collected. The supernatant (2 mL) were mixed with 2 mL of acid-ninhydrin solution (composed of 30 mL glacial acetic acid, 20 mL of 6 M phosphoric acid, and 1.25 g of ninhydrin) and an additional 2 mL of glacial acetic acid. The mixture was incubated at 100°C for 1 hour in a water bath. After incubation, the reaction tubes were immediately cooled on ice, and 4 mL of toluene was added to extract the chromophore. The mixture was vigorously vortexed for 15–20 seconds to facilitate phase separation. The upper toluene layer, containing the chromophore, was carefully separated from the aqueous phase and maintained at room temperature. Absorbance was measured at 520 nm using a UV spectrophotometer (Jin et al. 2017).

Statistical analysis

The data collected in this study were expressed as means $\pm$ standard deviation. A one-way analysis of variance (ANOVA) was performed to assess differences between sampling sites and tree species. In cases where statistical significance was found, post-hoc comparisons were conducted using Duncan's multiple range test ($p < 0.05$). All statistical analyses were performed using the SPSS program (ver. 24.0).

Results

Physiological characteristics

The diameter at breast height (DBH) of *P. yedoensis*

Table 3. DBH, soil moisture and vigor condition of *P×yedoensis* in Goyang and Paju, Gyeonggi Province, South Korea

Sites	DBH ^{ns} (cm)	Leaf area ^{***} (cm ²)	Leaf dry weight ^{***} (g)	Soil moisture ^{***} (%)	Vigor condition [*]
PY1	17.3±4.5 ^a	36.6±9.2 ^c	0.39±0.10 ^b	26.2±4.1 ^a	95.3±2.1 ^a
PY2	20.9±5.0 ^a	47.1±13.6 ^b	0.33±0.10 ^c	9.9±2.7 ^c	95.3±2.0 ^a
PY3	19.2±4.7 ^a	55.1±15.9 ^a	0.43±0.16 ^a	12.7±3.5 ^b	92.1±3.5 ^b

^{*}, ^{***}Data represent means±standard deviations in each column followed by the same letter did not differ significantly ($p < 0.05$, $p < 0.001$) by one-way analysis of variance (ANOVA) and Duncan's multiple range test.

PY1, Kintex-ro (Goyang); PY2, Soejae-ro (Paju); PY3, Mugunghwa-ro (Goyang).

ns, not significant.

Table 4. Changes of leaf net photosynthetic rate, transpiration rate, water use efficiency, stomatal conductance and sub stomatal CO₂ concentration of *P×yedoensis* in Goyang and Paju, Gyeonggi Province, South Korea

Sites	A ^{***}	E ^{***}	WUE ^{***}	g ^{**}	Ci ^{***}
PY1	18.1±1.7 ^a	3.2±0.3 ^a	5.7±0.4 ^b	0.3±0.1 ^a	292.8±15.2 ^b
PY2	16.2±1.8 ^b	2.9±0.5 ^b	5.7±0.9 ^b	0.2±0.1 ^b	278.8±24.7 ^c
PY3	15.8±3.3 ^b	2.0±0.3 ^c	8.2±1.6 ^a	0.3±0.1 ^a	332.7±33.7 ^a

^{**}, ^{***}Data represent means±standard deviations in each column followed by the same letter did not differ significantly ($p < 0.01$, $p < 0.001$) by one-way analysis of variance (ANOVA) and Duncan's multiple range test.

A, leaf net CO₂ assimilation rate ($\mu\text{mol CO}_2 \text{ m}^{-2} \cdot \text{s}^{-1}$); E, transpiration rate ($\text{mmol H}_2\text{O m}^{-2} \cdot \text{s}^{-1}$); WUE, water use efficiency ($\mu\text{mol} \cdot \text{mmol}^{-1}$); g, stomatal conductance ($\text{mmol H}_2\text{O m}^{-2} \cdot \text{s}^{-1}$); Ci, sub-stomatal CO₂ concentration ($\mu\text{mol CO}_2 \cdot \text{mol}^{-1}$).

PY1, Kintex-ro (Goyang); PY2, Soejae-ro (Paju); PY3, Mugunghwa-ro (Goyang).

Table 5. Chlorophyll fluorescence of *P×yedoensis* in Goyang and Paju, Gyeonggi Province, South Korea

Sites	Chlorophyll fluorescence		
	Fo [*]	Fm ^{**}	Fv/Fm [*]
PY1	175.1±21.0 ^a	930.0±120.5 ^a	0.81±0.03 ^c
PY2	134.2±11.6 ^c	922.6±54.3 ^{ab}	0.85±0.01 ^a
PY3	152.4±12.7 ^b	887.5±43.0 ^c	0.83±0.01 ^b

^{*}, ^{**}Data represent means±standard deviations in each column followed by the same letter did not differ significantly ($p < 0.05$, $p < 0.01$) by one-way analysis of variance (ANOVA) and Duncan's multiple range test.

PY1, Kintex-ro (Goyang); PY2, Soejae-ro (Paju); PY3, Mugunghwa-ro (Goyang).

street trees across the three study regions ranged from 17.3 to 20.9 cm, with no statistically significant differences among the sites (Table 3). In contrast, both leaf area and leaf dry weight showed significant regional variation. The highest values were observed in PY3, with a mean leaf area of $55.1 \pm 15.9 \text{ cm}^2$ and a dry weight of $0.43 \pm 0.16 \text{ g}$. Soil moisture content also differed significantly among the

study sites ($p < 0.001$), with PY1 exhibiting approximately 2.6 times higher moisture content compared to PY2. Although soil compaction in the street tree planting zones made it difficult to measure soil moisture accurately, clear differences among regions were observed. Tree vitality scores for *P×yedoensis* were consistently above 92 across all sites, indicating good overall health in all three regions.

Physiological characteristics

Photosynthesis and chlorophyll fluorescence response

To assess the physiological characteristics of *P×yedoensis* in urban environments, we measured photosynthetic efficiency, transpiration rate, water use efficiency, stomatal conductance, and intercellular CO₂ concentration (Table 4). The highest net CO₂ assimilation rate was observed at PY1, with an average of $18.1 \pm 1.7 \mu\text{mol CO}_2 \text{ mm}^{-2} \cdot \text{s}^{-1}$. Correspondingly, the transpiration rate at PY sites was also significantly elevated, with a value of $3.2 \pm 0.3 \text{ mmol H}_2\text{O m}^{-2} \cdot \text{s}^{-1}$ ($p < 0.001$). Water use efficiency was peaked at PY3, recording a mean of $8.2 \pm 1.6 \mu\text{mol CO}_2 \text{ mmol}^{-1} \text{ H}_2\text{O}$, which represents a 44% increase compared to

PY1 and PY2.

Intercellular CO₂ concentration was also significantly higher at PY3 ($332.7 \pm 33.7 \mu\text{mol CO}_2 \text{ mol}^{-1}$), approximately 1.1 to 1.2-fold greater than those recorded at PY1 and PY2, respectively ($p < 0.001$).

Chlorophyll fluorescence analysis revealed that the initial fluorescence (F₀) was lowest at PY2 (134.2 ± 11.6), while maximum fluorescence (F_m) did not significantly among sites. As a result, the maximum quantum yield of Photosystem II (F_v/F_m) was relatively high at PY2, with a value of 0.85 (Table 5).

Contents of chlorophyll and proline

Chlorophyll and carotenoid contents in *P. yedoensis* leaves collected from different survey sites are presented in Table 6. Chlorophyll concentrations remained relatively consistent among the sites, suggesting minimal variation in photosynthetic pigments associated with primary metabolic processes. In contrast, carotenoid content at PY1 was approximately 38% higher than at PY3.

Carotenoids are well known for their protective function under environmental stress, mitigating oxidative damage by quenching reactive oxygen species and stabilizing chlorophyll molecules. Therefore, the elevated carotenoid concentration at PY1 may reflect an adaptive physiological response to site-specific stress conditions.

To further assess plant stress responses, proline content was analyzed. Proline is an amino acid that accumulates under abiotic stresses conditions such as drought and salinity, serving as a biochemical marker of stress (Jin et al. 2017). The highest proline concentration was detected at PY1, a mean value of $3.76 \pm 1.24 \text{ mg} \cdot \text{g}^{-1}$ fresh weight (Fig. 1), which was significantly greater than those observed at other

sites ($p < 0.05$). These results suggest that *P. yedoensis* trees at PY1 experience more challenging growth conditions, which may impact their physiological stability and long-term health.

Discussion

Tree vitality, defined as the capacity of a tree to grow and sustain physiological function, is influenced by both natural and anthropogenic factors (Dobbertin 2005; Park et al. 2021b). Among the various methods available for assessing tree vitality, measuring the electrical resistance of the cambial layer provides a non-destructive and practical approach for evaluating tree health (Tattar and Blanchard 1976; Hwang and Kim 2016; Park et al. 2021b). In this study,

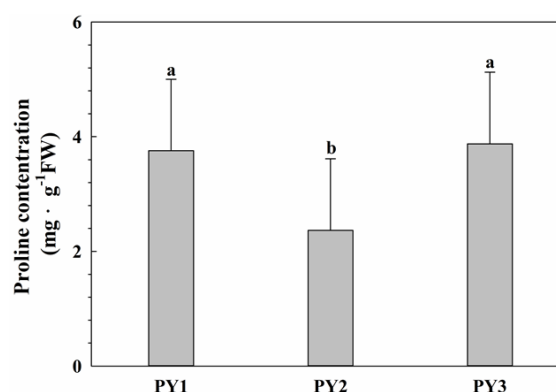


Fig. 1. Proline concentration of *P. yedoensis* in Goyang and Paju, Gyeonggi Province, South Korea. Error bars represent standard deviation. Columns marked with the same letter do not differ significantly at $p < 0.05$ according to one-way ANOVA followed by Duncan's multiple range test. PY1, Kintex-ro (Goyang); PY2, Soejae-ro (Paju); PY3, Mugunghwa-ro (Goyang).

Table 6. Chlorophyll contents of *P. yedoensis* in Goyang and Paju cities in Gyeonggi-do, Korea

Sites	Chlorophyll ^a	Chlorophyll ^b	Total Chlorophyll	Carotenoid
	(mg · g ⁻¹ FW)			
PY1	9.16 ± 3.67 ^a	3.48 ± 1.23 ^a	12.64 ± 4.90 ^a	1.85 ± 0.57 ^a
PY2	10.30 ± 2.60 ^a	4.16 ± 1.00 ^a	14.45 ± 3.57 ^a	1.54 ± 0.64 ^{ab}
PY3	9.24 ± 3.44 ^a	3.50 ± 1.07 ^a	12.74 ± 4.51 ^a	1.34 ± 0.19 ^b

Data represent Means ± standard deviations in each column followed by the same letter did not differ significantly ($p < 0.05$) by one-way analysis of variance (ANOVA) and Duncan's multiple range test.

PY1, Kintex-ro (Goyang); PY2, Soejae-ro (Paju); PY3, Mugunghwa-ro (Goyang).

tree vitality was assessed using the Junsmeter, a diagnostic instrument developed in Korea and modeled after the U.S.-made Shigometer. This device measures the electrical resistance of the cambial layer. The device provides values on a scale from 0 to 100, with higher values generally indicating better vitality. Readings above 87 are considered indicative of healthy tree growth. (Park et al. 2021b). All surveyed trees exhibited vitality values ranging from 92.1 to 95.3, confirming their generally healthy condition. This outcome is likely attributable to systematic street tree management by local governments since 2006, including regular maintenance practice such as irrigation and nutrient supplementation during periods of drought or environmental stress, which have contributed to sustaining high vitality levels.

At the PY1 site, soil moisture content was measured at 26.2%, and the net photosynthetic rate was $18.1 \pm 1.7 \mu\text{mol CO}_2 \text{ m}^{-2} \text{ s}^{-1}$, both higher than those observed at PY2 and PY3. Transpiration rates were also elevated at PY1, indicating active physiological processes supported by sufficient water availability. Conversely, PY2 and PY3 exhibited comparatively lower soil moisture and photosynthetic rates. Among the environmental factors influencing street tree health, soil water stress is a critical limiting factor that can inhibit growth and potentially cause mortality (Fini et al. 2009). Soil water deficits adversely affect leaf photosynthetic performance by inducing stomatal closure and altering the biochemical mechanisms underlying photosynthesis (Chaves and Oliveira 2004; Kagotani et al. 2016). Supporting this, Bhusal et al. (2021) reported decreases in both photosynthetic efficiency and leaf water potential in 1-year-old *Prunus sargentii* seedlings subjected to water supply reductions.

Interestingly, *P. yedoensis* trees at PY3 exhibited the highest intercellular CO_2 concentration ($332.7 \pm 33.7 \mu\text{mol CO}_2 \text{ mol}^{-1}$) and water use efficiency ($8.2 \pm 1.6 \mu\text{mol mmol}^{-1}$) among the three sites, despite showing the lowest photosynthetic rate ($15.8 \pm 3.3 \mu\text{mol CO}_2 \text{ m}^{-2} \text{ s}^{-1}$) and transpiration rate ($2.0 \pm 0.3 \text{ mmol H}_2\text{O m}^{-2} \text{ s}^{-1}$). This pattern suggests that although stomata were partially open, photosynthetic activity was suppressed, likely as a physiological response to water stress. Such a response is commonly observed under drought conditions, where plants reduce water loss by limiting stomatal opening while maintaining

minimal gas exchange (Je et al. 2006). Under moderate drought stress, stomatal conductance decreases, leading to reductions in both net photosynthesis and transpiration; however, because transpiration is more strongly inhibited than photosynthesis, water use efficiency typically increases (Xu et al. 2010).

Conversely, at PY2, photosynthesis and transpiration occurred at moderate levels, but both intercellular CO_2 concentration and water use efficiency were relatively low. This indicates that stomatal closure effectively limited CO_2 uptake to reduce water loss, which consequently compromised photosynthetic performance.

Climate change, including the urban heat island effect, directly and indirectly influences plant growth, potentially causing damage to photosynthetic apparatuses such as Photosystem II (PSII) (Takahashi and Murata 2008). Chlorophyll fluorescence analysis is widely employed to evaluate PSII responses, offering valuable insights into plants' physiological reactions to environmental stress and overall photosynthetic performance (Strasser and Strasser 1995). International studies report that healthy leaves typically exhibit an Fv/Fm value near 0.83 (Johnson et al. 1993; Maxwell and Johnson 2000), whereas exposure to stressors often results in a decline in this parameter (Kim et al. 2018; Kycko et al. 2018).

Chlorophyll, an essential pigment for photosynthesis, was measured in this study, with total chlorophyll content ranging from 12.64 to 14.45 mg g^{-1} fresh weight and showing no significant differences across sites. Generally, environmental stressors such as drought, ozone exposure, and soil pollution have been reported to degrade chlorophyll and reduce its concentration in plant tissues. For example, Cho et al. (2020) documented up to a 43% decrease in chlorophyll content in 2-year-old street tree seedlings subjected to stress treatments compared to controls. Similarly, Bhusal et al. (2021) observed reduced chlorophyll content in *Prunus sargentii* seedlings experiencing water deficiency. Such stress-induced chlorophyll degradation often correlates with diminished photosynthetic activity and visible symptoms such as leaf yellowing and chlorosis (Sung et al. 2010).

Conversely, some studies have found that chlorophyll content may increase in plants exhibiting high tolerance to pollutants (Chaudhary and Rathore 2019), with elevated

chlorophyll levels potentially enhancing resistance to environmental contaminants (Missanjo et al. 2015). Moreover, Lee et al. (2022a) reported that leaves subjected to water deficit conditions may exhibit relatively higher chlorophyll concentrations. The lack of significant variation in chlorophyll content among sites in this study suggests minimal differences in stress factors such as air pollution or water availability, or alternatively, that the studied *P. × yedoensis* trees have developed adaptive tolerance to urban stressors including drought and pollution.

Proline is a multifunctional amino acid that acts as a compatible osmolyte, playing critical roles in osmoregulation, membrane stabilization, and metabolic support under environmental stress conditions (Kishor et al. 1995; Peng et al. 1996; Bartels and Sunkar 2005; Jin et al. 2017). It is well documented that proline accumulation increases under environmental stress, making it a useful physiological indicator for evaluating plant stress responses and tolerance (Abrahám et al. 2010). Plants synthesize proline to mitigate stress damage, contributing to osmotic adjustment and maintenance of photosynthetic efficiency (Reddy et al. 2015). Accordingly, elevated proline content is commonly interpreted as an adaptive response to water-deficit stress (Liu and Zhu 1997; Jin et al. 2017; Lee et al. 2023). As an osmoprotectant, proline concentration increases under drought conditions to regulate osmotic pressure and enable adaptation to water-limited environments (Kijowska-Oberc et al. 2023).

In this study, proline contents at PY1 (3.76 mg g⁻¹ FW) and PY3 (3.88 mg g⁻¹ FW) were significantly higher than at PY2, indicating a greater degree of physiological stress. This pattern can be contextualized by photosynthetic parameters: although soil moisture at PY1 was higher than at other sites, the actual water available for uptake may have been limited, potentially leading to lower water use efficiency relative to photosynthesis and transpiration rates—possibly due to stress induced by transpiration or inefficient water utilization. Conversely, PY3 exhibited lower transpiration relative to photosynthesis, indicating higher water use efficiency, which was accompanied by significant proline accumulation as a physiological adaptation to water stress. Meanwhile, PY2's relatively low proline content suggests a lower level of water stress or a diminished stress response at that site. However, because standard reference

values for proline content in urban street trees have yet to be established, further long-term studies and continuous monitoring are necessary to develop baseline thresholds and deepen our understanding of proline dynamics under urban environmental stresses.

Comparatively, Lee et al. (2023) reported that *Zelkova serrata* street trees in northern Gyeonggi exhibited lower photosynthetic efficiency and maximum photochemical efficiency (Fv/Fm) than *Prunus* spp. The observed differences likely reflect inherent physiological traits unique to each species, underscoring the need for species-specific strategies to mitigate urban environmental stress. Consequently, refining species selection and planting strategies is crucial to enhance the sustainability of urban ecosystem services and effectively address the challenges posed by future climate change.

Conclusion

This study examined the physiological characteristics of *P. × yedoensis*, a key urban tree species, in the context of climate change, urban heat island effects, and air pollution. Urban environments often present challenging conditions for tree growth due to factors such as air pollution and sub-optimal light availability, which can induce physiological stress. However, our analyses of photosynthetic activity, tree vitality, chlorophyll content, and proline accumulation suggest that these street trees exhibit relatively high stress tolerance. Nonetheless, the absence of standardized stress tolerance indices for major urban tree species highlights the need for further comprehensive research to accurately evaluate tree responses under urban environmental pressures. Moreover, with the increasing role of local governments in urban forest management, implementing long-term monitoring programs that integrate both growth and physiological assessments, alongside effective management practices, is essential. Such strategies will underpin the sustainable and resilient development of urban forests, ensuring their continued health and ecological functions amidst ongoing environmental challenges.

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